

tude of the voltage applied to the input of multiplier chain. Also, in both cases, multiplied output voltage is equal to nV_p , provided the bracketed term in the denominator is zero or negligible.

This is possible for large value of capacitance C.

Another voltage multiplier circuit that offers a superior performance is the one that uses unequal capacitors.

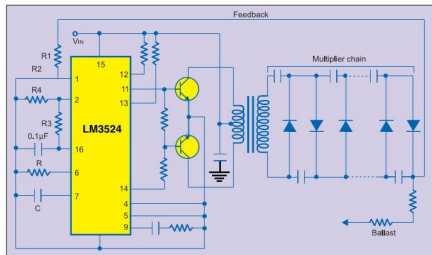


Fig. 7: Push-pull converter with cascaded multiplier chain

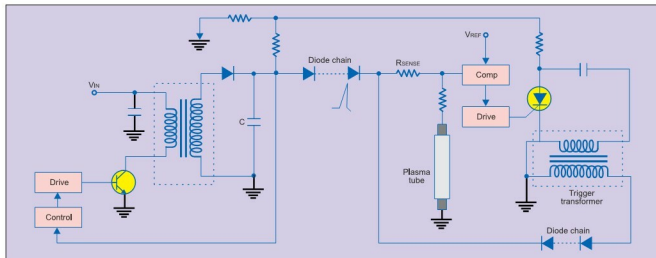


Fig. 8: Fly-back converter with high-voltage trigger circuit

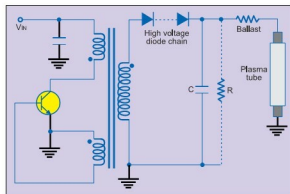


Fig. 9: Constant power output fly-back converter

capacitor type voltage multiplier chain, as shown in Fig. 7. The push-pull circuit generates a high-frequency square waveform across transformer secondary, which is then multiplied to generate the starting high voltage for the discharge tube. The push-pull circuit here is configured around a switching control IC type LM3524.

It may be mentioned here that, a large number of control ICs intended for switching power supply design are commercially available. Some of the other popular type numbers include TL497, TL594 and TDA8130. Their data sheets contain typical application circuits for switching supply design.

When discharge occurs, the voltage falls to a lower value to supply the sustaining discharge current through the ballast resistance. The rest of the voltage is dropped across the multiplier chain components. In some cases, the

Fig. 6 shows the circuit configuration.

Helium-neon power supply circuits

The most commonly used helium-neon laser power supply circuit is a push-pull type high-frequency switched mode DC-to-AC power supply cascaded to a diode-

sustaining voltage may be tapped from an earlier point in the chain for better overall performance. In that case, the output of the multiplier chain feeds the discharge tube through a relatively high resistance of the order of tens of megaohms and the sustaining voltage feeds the tube through the ballast resistance.

The other common circuit topology is the one where the power supply generates the voltage required to sustain the discharge and a high-voltage trigger circuit generates the starting